Double Integrals

ANSWER KEY

Section 1: Iterated Integrals over Rectangles

1 Evaluate $\int_0^2 \int_0^3 (xy^2)\,dx\,dy$.

a. 12 ✓

b. 18

c. 9

d. 6

Factor: $\left(\int_0^3 x\,dx\right)\left(\int_0^2 y^2\,dy\right) = [x^2/2]_0^3 [y^3/3]_0^2 = (9/2)(8/3) = 72/6 = 12.$

2 Evaluate $\int_0^1 \int_0^2 (2x + 3y)\,dy\,dx$.

a. 8 ✓

b. 6

c. 10

d. 4

Inner: $\int_0^2 (2x + 3y)\,dy = [2xy + \tfrac{3}{2}y^2]_0^2 = 4x + 6$. Outer: $\int_0^1 (4x + 6)\,dx = [2x^2 + 6x]_0^1 = 2 + 6 = 8.$

3 What does Fubini's Theorem guarantee for a continuous function $f(x, y)$ over a rectangle $R = [a, b] \times [c, d]$?

a. $\int_a^b \int_c^d f\,dy\,dx = \int_c^d \int_a^b f\,dx\,dy$ ✓

b. The integral equals the perimeter times height

c. The order of integration cannot be swapped

d. The integral must be zero

Fubini's Theorem ensures that the order of integration can be reversed without changing the value: $\iint_R f\,dA = \int_a^b \int_c^d f\,dy\,dx = \int_c^d \int_a^b f\,dx\,dy$.

Section 2: Double Integrals over General Regions

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4 Evaluate $\iint_D (x+y)\,dA$ where D is bounded by $y=0$, $y=x$, and $x=1$.

a. $1/2$ ✓

b. $1/3$

c. $2/3$

d. $1/6$

$\int_0^1 \int_0^x (x+y)\,dy\,dx = \int_0^1 [xy + y^2/2]_0^x\,dx = \int_0^1 (x^2 + x^2/2)\,dx = \int_0^1 \frac{3}{2}x^2\,dx = [\frac{x^3}{2}]_0^1 = 1/2.$

5 Reverse the order of integration for $\int_0^1 \int_x^1 f(x,y)\,dy\,dx$.

a. $\int_0^1 \int_0^y f(x,y)\,dx\,dy$ ✓

b. $\int_0^1 \int_y^1 f(x,y)\,dx\,dy$

c. $\int_0^1 \int_0^1 f(x,y)\,dx\,dy$

d. $\int_x^1 \int_0^1 f(x,y)\,dx\,dy$

The region is $0 < x < 1$ and $x \le y \le 1$. In terms of y: $0 \le y \le 1$, and for each y, x ranges from 0 to y. Thus $\int_0^1 \int_0^y f(x,y)\,dx\,dy$.

6 Evaluate $\int_0^1 \int_y^1 e^{x^2}\,dx\,dy$ by reversing the order of integration.

a. $\frac{e-1}{2}$ ✓

b. $e-1$

c. $\frac{e}{2}$

d. $e^2 - 1$

Reversed: $\int_0^1 \int_0^x e^{x^2}\,dy\,dx = \int_0^1 x e^{x^2}\,dx = [\frac{1}{2}e^{x^2}]_0^1 = \frac{e-1}{2}$.

Section 3: Double Integrals in Polar Coordinates

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7 What is the area element dA in polar coordinates?

a. $dA = r\,dr\,d\theta$ ✓

b. $dA = dr\,d\theta$

c. $dA = r^2\,dr\,d\theta$

d. $dA = \frac{1}{r}\,dr\,d\theta$

The Jacobian factor for polar coordinates is r, so the infinitesimal area element is $dA = r\,dr\,d\theta$.

8 Evaluate $\iint_R (x^2 + y^2)\,dA$ over the disk $x^2 + y^2 \le 4$.

a. 8π ✓

b. 16π

c. 4π

d. 32π

In polar: $x^2 + y^2 = r^2$. Bounds: $0 \le r \le 2$, $0 \le \theta \le 2\pi$.
$\int_0^{2\pi} \int_0^2 r^2 (r\,dr)\,d\theta = 2\pi \int_0^2 r^3\,dr = 2\pi[r^4/4]_0^2 = 2\pi(4) = 8\pi.$

9 Evaluate the Gaussian integral $\int_{-\infty}^{\infty} e^{-x^2}\,dx$ using polar coordinates.

a. $\sqrt{\pi}$ ✓

b. π

c. $2\sqrt{\pi}$

d. $\pi/2$

$I^2 = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)} dA = \int_0^{2\pi} d\theta \int_0^{\infty} r e^{-r^2} dr = 2\pi[-\frac{1}{2}e^{-r^2}]_0^{\infty} = 2\pi(1/2) = \pi \Rightarrow I = \sqrt{\pi}.$

Section 4: Volume, Mass, and Center of Mass

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10 Find the volume under the paraboloid $z = 4 - x^2 - y^2$ above the xy-plane.

a. 8π ✓

b. 16π

c. 4π

d. 2π

Region D: disk of radius $r = 2$. $V = \int_0^{2\pi}\int_0^2 (4 - r^2) r\,dr\,d\theta = 2\pi[2r^2 - r^4/4]_0^2 = 2\pi(8 - 4) = 8\pi$.

11 A lamina occupies the region $[0, 1] \times [0, 2]$ with density $\rho(x, y) = 3x$. Find its total mass M.

a. 3 ✓

b. 6

c. 1.5

d. 4.5

$M = \int_0^1\int_0^2 3x\,dy\,dx = 2\int_0^1 3x\,dx = 2[\tfrac{3}{2}x^2]_0^1 = 3$.

12 State the formula for the center of mass $\bar{x}$ of a lamina with density $\rho(x, y)$ and total mass M.

a. $\bar{x} = \frac{1}{M}\iint_D x\rho(x, y)\,dA$ ✓

b. $\bar{x} = \frac{1}{M}\iint_D y\rho(x, y)\,dA$

c. $\bar{x} = \iint_D x\,dA$

d. $\bar{x} = \frac{M}{\iint x\,dA}$

The x-coordinate of the center of mass is the moment about the y-axis divided by total mass: $\bar{x} = \frac{M_y}{M} = \frac{1}{M}\iint_D x\rho(x, y)\,dA$.

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